

Physics-Informed Neural Networks applied to predict Bending Moment of circular tunnels using observed displacements

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1 Introduction

The behaviour and structural design of underground structures is governed by the distribution of internal forces. Out of these internal forces, bending moments are most critical for structures supporting bending forces. In tunnels, axial forces are equally relevant, not for reinforcement considerations only, but to guarantee its stability as well. However, despite the importance of these internal forces, traditional monitoring techniques of these structures concentrate on measuring total or relative deformations to verify design assumptions rather than enabling direct conclusions about the governing internal forces of the structure itself. (Fuentes [2015]).

With regards to bending moments, the available solutions to obtain bending moments and axial loads from displacements involve back-calculation and iterative processes using models that are successful in forward prediction – e.g., continuum models, or finite element analysis.

2 Objective

In this case, we want to use physics Informed Neural Networks to predict bending moments of circular tunnels.

Here, the focus is on circular tunnels for simplicity, although the same principles apply to other section shapes. It is assumed that the tunnel lining structure is monolithic (i.e. the joints are not articulated) and provides full structure continuity.

Radial displacements, perpendicular to the tunnel cross section, were used (see Fig. 1). However, the observed displacements (u_O) can be divided into two categories: those that produce bending moments and those that do not. The latter, in practice, are rigid body (u_{RB}) displacements such as translations and rotations, and uniform convergence (u_C) displacements – i.e., a uniform reduction in the tunnel diameter (see Fig. 1). The former are displacements referred here as distortion displacements (u_D).

3 Governing Equations

Most of the widely accepted analytical and widespread used solutions for tunnel lining design define the shape of bending moments and radial displacements in tunnels using multiples of the cosine – e.g. $\cos(2u)$. The equation linking radial displacement and bending moment of a circular beam of thin section can be calculated as follow (for more details about these governing equations, the readers should study the reference paper, e.g. Fuentes [2015]).

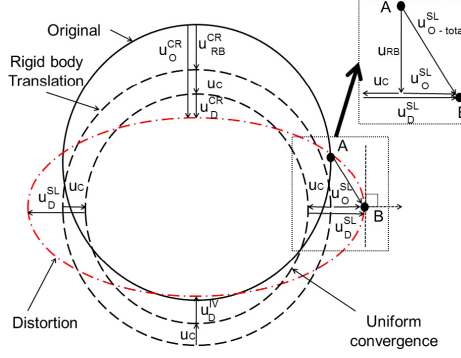


Figure 1: Tunnel displacements definitions.

- The Radial displacements can be calculated as:

$$u_D = a + b \cos \phi + c \cos 2\phi \quad (1)$$

where for case of Carranza-Torres et al. [2013], the coefficients of the radial displacements function are $a = -0.0009717$, $b = 0.0003633$, and $c = 0.01539$ respectively.

Also for case of Brinkgreve et al. [2011], the coefficients of the radial displacements function are $a = 0.0007949$, $b = 0.0001159$, $c = -0.0004788$ respectively.

-The partial deferential equation (PDE) linking radial displacement and bending moment is:

$$\frac{d^2 u_D}{d\phi^2} + u_D = -\frac{R^2 M(\phi)}{EI} \quad (2)$$

where R is the radius of the tunnel equal to 5 m, and EI is 26,0416.7 kN/m^2 for case of Carranza-Torres et al. [2013].

4 Tasks

- Build a Physics-Informed Neural Network (PINNs) by considering the governing equations and proper boundary conditions.
- Tune the parameters and hyper-parameters of the PINNs in order to achieve a forward training.
- Compare the results to the true solutions.
- Build another PINNs based on the true solutions in order to have a inverse training.
- Compare the results and calculate the errors and try to minimize the error by tuning the parameters and hyper-parameters of the PINNs.
- Provide a short presentation.

References

Raul Fuentes. Internal forces of underground structures from observed displacements. *Tunnelling and underground space technology*, 49:50–66, 2015.

C Carranza-Torres, B Rysdahl, and M Kasim. On the elastic analysis of a circular lined tunnel considering the delayed installation of the support. *International Journal of Rock Mechanics and Mining Sciences*, 61:57–85, 2013.

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